PART-A

Q1). Represent each students subject scores as a vector

→ A vector is an ordered list of numbers representing magnitude and direction in space. Each student's marks ~~are~~ across subjects can be represented as a vector in multidimensional space

~~Q2~~ Vector For:

$$S1 : [88, 74, 91, 67, 82]$$

Q2). a) Compute Norm-1 and Norm-2 of vectors

→ L1 Norm (Manhattan norm)

$$\|v\|_1 = |x_1| + |x_2| + \dots + |x_n|$$

For S_1:

$$\|S_1\|_1 = 88 + 74 + 91 + 67 + 82$$

$$= 402$$

L2 Norm (Euclidean Norm)

$$\|v\|_2^2 = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

$$\|S_1\|_2 = \sqrt{68^2 + 74^2 + 91^2 + 67^2 + 82^2}$$

$$= \sqrt{7744 + 5476 + 8281 + 4489 + 6724}$$

$$= \sqrt{32714}$$

$$= 181.0$$

Q2)

b) Dot Product & Angle Between Two Students

Dot product measures similarity between two variables

Eg b/w S_1 and S_2:

$$A \cdot B = A_1B_1 + A_2B_2 + \dots + A_nB_n$$

Eg:

Angle b/w vectors

$$\cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|}$$

c). Cross Product (3D subjects)

-) Cross product works only for 3 dimensions, so we use:

- Math
- Physics
- Chemistry

Theory

Example:

$S_1 = [88, 74, 91]$
$S_2 = [78, 69, 73]$

$$A \times B = \begin{vmatrix} i & j & k \\ 88 & 74 & 91 \\ 78 & 69 & 73 \end{vmatrix}$$

Q3). Projection of one vector onto Another

-) Projection shows how much one vector lies in direction of another.

Formula:

$$\text{proj}_B(A) = \frac{A \cdot B}{\|B\|^2} B$$

~~Q42 From St~~

Q5) Line, Plane, Hyperplane

-) Line

1- Dimensional relationship

Eg:

Math v/s Physics

$$y = mx + c$$

2- Dimensional surface in 3D

$$ax + by + cz + d = 0$$

Eg:

Math, Physics, Chemistry

Hyperplane

- Higher dimensional generalization

For 5 subjects

$$w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + w_5 x_5 + b = 0$$

Q6). Dimensional increases

Dimension	Example
2D	Math vs Physics
3D	Math Physics Chemistry
5D	All Subjects

Higher dimensions from hyperplanes

Q7). Eigenvalues & Eigenvectors of Covariance Matrix

Theory

Eigenvectors shows principal directions of variance.

Steps

1. Compute covariance matrix
2. Solve eigenvalue equation

$$Av = \lambda v$$

(Q8) LU Decomposition

→ Theory

Matrix decomposition:

$$A = LU$$

L → lower triangular
U → upper triangular

(Q9) Singular Value Decomposition

→ $A = U \Sigma V^T$

(Q10) PCA

→ Principal Component Analysis reduces features while keeping maximum variance

Steps

1. Standardize data
2. Covariance matrix
3. Eigen decomposition
4. Select top components

(Q11). Linear Discriminant Analysis

→ LDA seperate classes

Classes in dataset:
- Above Average
- Below Average